# Proper Scoring Rules and Calibration

## ELG 5218 - Uncertainty Evaluation in Engineering Measurements and Machine Learning

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March 3, 2026

# Outline: Proper Scoring Rules and Calibration

- **Motivation & Definitions**
- **Proper Scoring Rules:** NLL / Brier / CRPS
- **Calibration Diagnostics:** reliability diagram, ECE
- **Empirical Findings:** modern deep nets are miscalibrated
- **Calibration Methods:** temperature scaling, Platt, histogram, isotonic
- **Applications & Notebooks:** image classification, detection, LLMs
- **Case Study & Code Demo**
- **Wrap-up & Reading/Assignments**

*Next lecture: conformal prediction (finite-sample coverage, efficiency).*

# Why do we care about *probabilities*?

- Many decisions need **risk-aware** predictions (medicine, autonomy, finance).
- **Accuracy is not enough:** two models with identical accuracy can have very different uncertainty quality.
- We want a metric that rewards:
  - **Calibration:** predicted probabilities match empirical frequencies.
  - **Sharpness:** confident when correct, uncertain when unsure.

Key question: how do we score a *predictive distribution* $p_\theta(y \mid x)$?

# Scoring rules (setup)

We observe $(x, y)$ and a model outputs a predictive distribution $p_\theta(y \mid x)$.

A **scoring rule** assigns a number $S(p_\theta, (x, y))$.

- Higher score (or lower loss) means better probabilistic prediction.
- We care about the **expected** score under the true data-generating process.

**Proper scoring rule:** the expected score is optimized when the predictive distribution equals the true conditional distribution:

$$\mathbb{E}_{(x,y)\sim p^*}\left[S(p_\theta, (x, y))\right] \leq \mathbb{E}_{(x,y)\sim p^*}\left[S(p^*, (x, y))\right]$$

(with equality iff $p_\theta(y \mid x) = p^*(y \mid x)$).

Reference: Murphy, *Probabilistic Machine Learning: Advanced Topics*, Sec. 14.2.1 (uploaded excerpt).

# Binary intuition: why "proper" matters

Binary event $y \in \{0, 1\}$. Suppose the true probability is $q = \mathbb{P}(y = 1)$, and the model reports $p$.

**Log score (NLL):**

$$\ell_{\text{NLL}}(p; y) = -\left[y \log p + (1 - y) \log(1 - p)\right]$$

Then the expected loss is

$$\mathbb{E}[\ell_{\text{NLL}}(p; y)] = -\left[q \log p + (1 - q) \log(1 - p)\right]$$

Taking derivative and setting to zero gives optimum at $p = q$ (strictly proper).

**Key takeaway:** proper scores discourage "hedging" or systematically over/under-confident probabilities.

# NLL / Log loss (classification)

**Negative log-likelihood (NLL)** is the standard proper scoring rule for probabilistic classifiers:

$$\ell_{\text{NLL}} = -\log p_\theta(y \mid x)$$

- Equivalent to cross-entropy for one-hot labels.
- Strongly penalizes assigning tiny probability to the true label ("tail sensitive").
- Minimizing NLL is aligned with maximizing likelihood / minimizing KL to the true distribution.

In Murphy's text, the log-likelihood is presented as a proper scoring rule and linked to Gibbs' inequality.

# Brier score (classification)

For $C$-class classification with probabilities $\hat{p}_c = p_\theta(y = c \mid x)$ and one-hot label $\mathbb{I}(y = c)$,

$$\text{Brier} = \frac{1}{C} \sum_{c=1}^{C} (\hat{p}_c - \mathbb{I}(y = c))^2$$

- Proper scoring rule.
- Less sensitive than NLL to extreme probabilities.
- Interpretable as squared error between predicted probability vector and one-hot label.

Original: Brier (1950). Modern ML discussion in Murphy, Sec. 14.2.1.

# CRPS (regression / continuous outcomes)

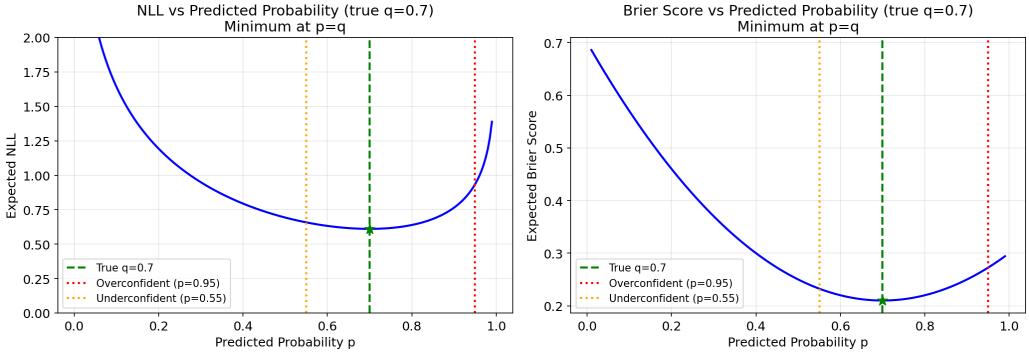
For continuous $y$ and predictive CDF $F$, the **continuous ranked probability score (CRPS)** is:

$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty} \left(F(z) - \mathbb{I}(z \geq y)\right)^2 dz$$

- Strictly proper scoring rule for distributions on $\mathbb{R}$.
- Generalizes absolute error from points to distributions.
- Useful when we predict full distributions (e.g., Gaussian mean+variance) or quantiles.

Theory: Gneiting & Raftery (JASA, 2007). Practical implementations exist in many libraries.

# Visual: NLL vs Brier penalties (binary)




**Interpretation:** NLL explodes as $p \to 0$ when $y = 1$ (or $p \to 1$ when $y = 0$). Brier grows quadratically.

# Calibration: definition

A probabilistic classifier is **calibrated** if:

$$\mathbb{P}(y = 1 \mid \hat{p} = 0.9) \approx 0.9$$

More generally, for multiclass, the predicted confidence should match empirical accuracy.

- Calibration is about **probabilities**, not just class predictions.
- Proper scoring rules (like NLL) encourage calibration in theory, but modern deep nets can still be miscalibrated.

Murphy, Sec. 14.2.2 (uploaded excerpt).

# Reliability diagram + ECE (Expected Calibration Error)

**Reliability diagram:** bin predictions by confidence; plot (bin accuracy) vs (bin confidence).

With $B$ bins, let $B_b$ be indices whose confidence falls in bin $b$.

$$\mathrm{acc}(B_b) = \frac{1}{|B_b|} \sum_{i \in B_b} \mathbb{I}(\hat{y}_i = y_i), \qquad \mathrm{conf}(B_b) = \frac{1}{|B_b|} \sum_{i \in B_b} \hat{p}_i$$

**ECE:**

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|B_b|}{N} \left|\mathrm{acc}(B_b) - \mathrm{conf}(B_b)\right|$$

ECE and reliability diagrams summarized in Murphy, Sec. 14.2.2.1 (uploaded excerpt).

# Notes on ECE (what to tell students)

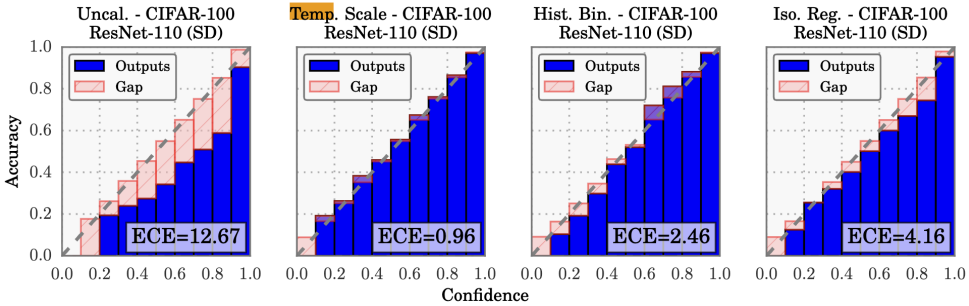
- ECE depends on **binning** choices (uniform bins vs equal-frequency bins).
- ECE often uses only the **top-1 confidence** in multiclass settings; this can miss miscalibration of non-top classes.
- Alternatives:
  - classwise / marginal calibration error,
  - adaptive binning,
  - differentiable calibration losses (for training-time objectives).

# Modern deep nets: accurate but miscalibrated

Empirical result (Guo et al., ICML 2017):

- Modern architectures (ResNet, DenseNet, etc.) are often **overconfident**.
- Depth, width, weight decay, and BatchNorm can **worsen** calibration.
- A simple post-hoc fix, **temperature scaling**, is surprisingly effective.

# Visual: Temperature Scaling




Source: "On Calibration of Modern Neural Networks" (ICML 2017).

# Post-hoc calibration: Platt vs Temperature scaling

Let $\boldsymbol{z}$ be logits from a trained classifier (fixed model). We learn a small calibration map on a validation set.

**Platt scaling (binary):** $q = \sigma(az + b)$
**Temperature scaling (multiclass):**

$$q = \mathrm{softmax}(\boldsymbol{z}/T)$$

- Single parameter $T > 0$ fitted by minimizing NLL on a calibration set.
- Does *not* change the predicted class (argmax unchanged).
- Often strong performance and very stable.

Murphy, Sec. 14.2.2.5; Guo et al., ICML 2017.

# Nonparametric calibration: histogram binning & isotonic regression

**Histogram binning:**

- Partition predicted confidences into bins.
- Replace predictions in each bin by the empirical accuracy in that bin.

**Isotonic regression:**

- Learn a monotone non-decreasing function $q = f(p)$ from validation data.
- More flexible than Platt scaling, but can overfit with small calibration sets.

Murphy, Sec. 14.2.2.4.

# Practical guidance (what to do in projects)

- Always keep a **calibration set** separate from training data.
- Report both:
  - **discrimination** (accuracy / AUROC),
  - **calibration** (reliability plot, ECE, NLL, Brier).
- Post-hoc methods are cheap and easy; temperature scaling is a strong default.
- Under distribution shift, calibration can degrade (monitor on new domains).

# Applications: same idea, different outputs

Calibration tools are modular: you need **(confidence, correctness)** pairs.

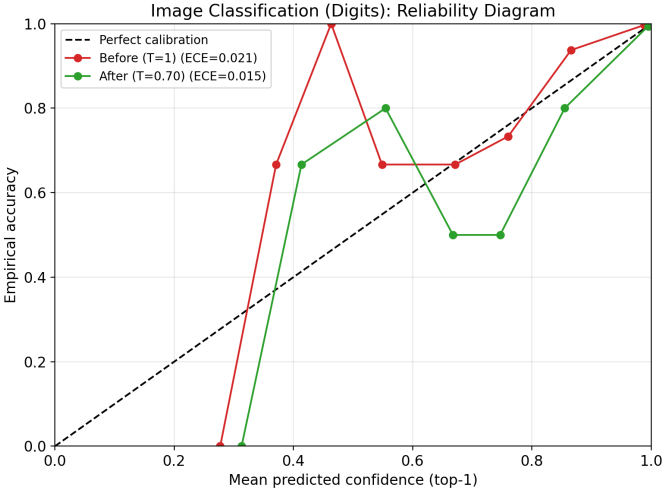
- **Image classification:** confidence = top-1 probability; correctness = (top-1 is correct).
- **Object detection:** confidence = detection score; correctness = (matched to GT with correct class + IoU$\geq \tau$).
- **LLMs/VLMs:** confidence = "probability answer is correct" (self-reported or derived); correctness = evaluation label.

**Same diagnostics:** reliability diagram, ECE, NLL/Brier (binary), NLL (multiclass).

# Image classification example: temperature scaling (Digits)

- train a multinomial classifier on `digits`,
- fit a single temperature $T$ on a calibration split by minimizing NLL,
- show reliability diagram + ECE before/after.

# Image classification example: temperature scaling (Digits)



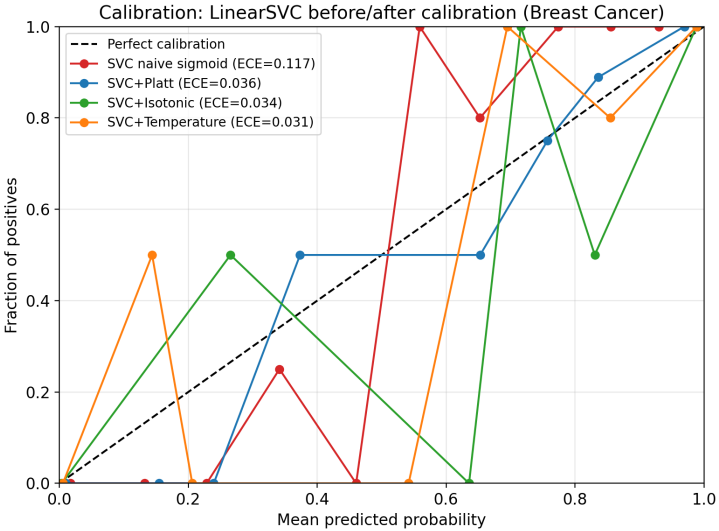
# Hands-on notebooks

Two practical resources to base your in-class demo and lab on:

- scikit-learn calibration example: `plot_calibration_curve`
- CalibrationTutorial (notebooks + explanations): `luferrer/CalibrationTutorial`

**Course notebook (this lecture):** `calibration_lab_notebook.ipynb`

# Hands-on notebooks



# Object detection: what is "confidence calibration"?

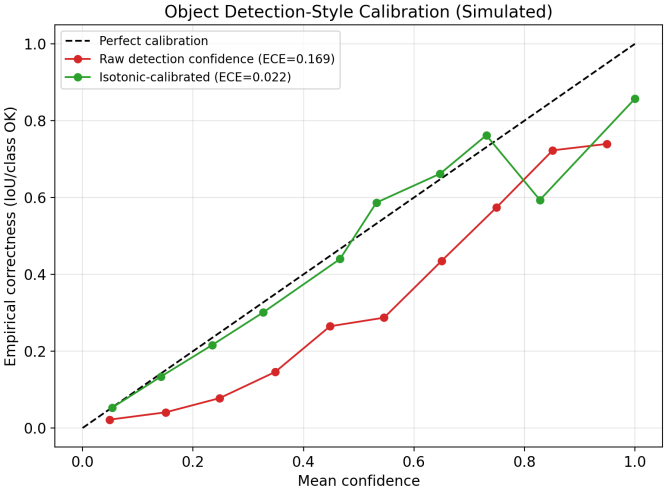
Detection confidence is often treated as **a probability of correctness**, but:

- it is not optimized for calibration (it is tuned for ranking/AP),
- it may shift across domains (weather, viewpoint, sensor, lighting),
- it may vary by class, size, occlusion, and localization quality.

A common evaluation protocol:

- label each detection as correct/incorrect using matching (class + IoU),
- plot reliability diagram of confidence vs correctness,
- calibrate using Platt/isotonic/temperature (or class-conditional variants).

# Object detection: what is "confidence calibration"?



# LLMs: confidence, abstention, and calibration

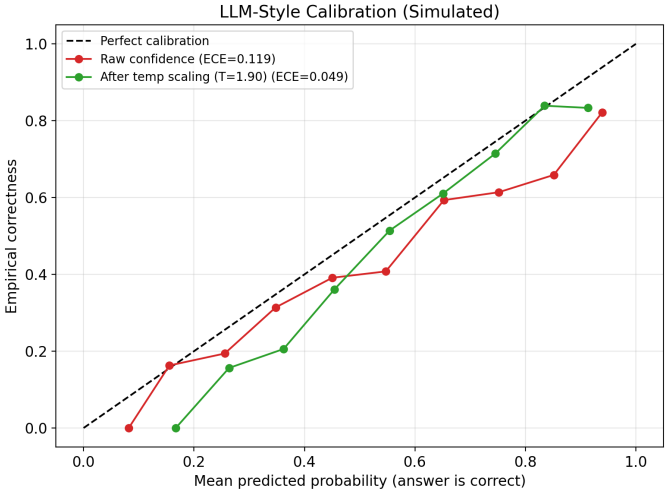
Typical setup:

- model answers a question and outputs a confidence score (explicit or derived),
- we evaluate correctness against a benchmark,
- calibrate confidence to avoid overconfident errors ("alarm fatigue" / unsafe deployment).

Practical knobs:

- **post-hoc calibration:** temperature scaling or isotonic on (confidence, correctness),
- **selective prediction:** abstain when confidence is low (ties to conformal in next lecture).

# LLMs: confidence, abstention, and calibration



# Assignment

**Goal:** reproduce + extend calibration analysis.

1. Pick one domain (image classification / detection / LLM QA).
2. Compute: NLL, Brier (if binary), reliability diagram, ECE (report binning choice).
3. Apply multiple calibration methods.
4. Report: calibration improvement *and* any change in accuracy / AP / selective risk.

**Stretch:** analyze calibration under a domain shift (corruptions, weather, prompt change).

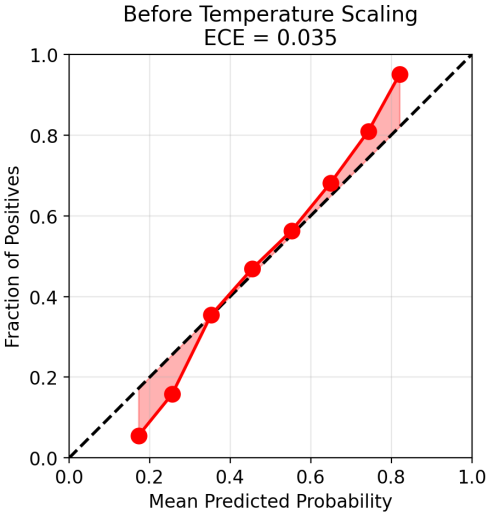
# Toy demo: miscalibration + temperature scaling

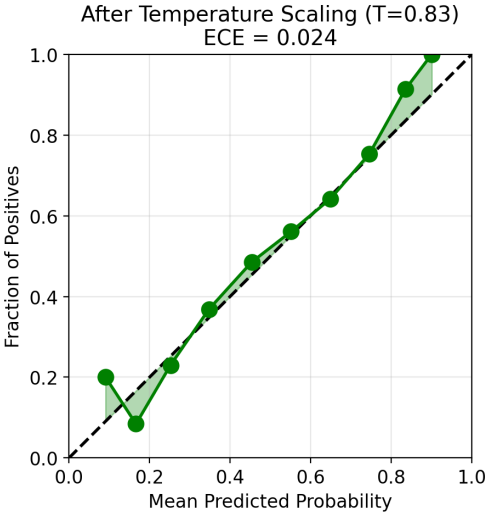
We simulate an **overconfident** classifier and then learn $T$ on a calibration split by minimizing NLL.

- Before: high ECE (overconfident probabilities).
- After: much lower ECE and improved NLL / Brier.

The next slide shows reliability diagrams from a small Python demo (provided with this deck).

# Reliability diagrams (toy example)



# Toy demo metrics (test set)

| Metric | Before | After (Temp. scaling) |
|---|---|---|
| NLL | 0.603 | 0.598 |
| Brier | 0.208 | 0.206 |
| ECE (10 bins) | 0.035 | 0.024 |

In this run, learned temperature $T \approx 0.83$ (higher $T$ softens overconfident logits).

# Wrap-up: key takeaways

- Use **proper scoring rules** (NLL, Brier, CRPS) to evaluate *probabilistic* predictions.
- Calibration is about **matching probabilities to frequencies**.
- Diagnose with **reliability diagrams** and summary metrics (ECE + NLL/Brier).
- Practical default: **temperature scaling** is simple and often very effective.

**Next lecture:** conformal prediction (finite-sample coverage + efficiency trade-offs).

# References


- Kevin P. Murphy. *Probabilistic Machine Learning: Advanced Topics*. (Calibration + proper scoring rules, Sec. 14.2).
- Chuan Guo, Geoff Pleiss, Yu Sun, Kilian Q. Weinberger. *On Calibration of Modern Neural Networks*. ICML 2017.
- Tilmann Gneiting, Adrian E. Raftery. *Strictly Proper Scoring Rules, Prediction, and Estimation*. JASA 2007.
- Glenn W. Brier. *Verification of Forecasts Expressed in Terms of Probability*. Monthly Weather Review 1950.
- J. Oberdiek et al. *Estimating and Evaluating Regression Predictive Uncertainty in Deep Object Detectors*. arXiv:2101.05036.
- L. Küppers et al. *Multivariate Confidence Calibration for Object Detection*. CVPR Workshops 2020.
- (LLMs) Example reading: *Language Models (Mostly) Know What They Know* (confidence estimation for LMs).
- Practical notebooks: scikit-learn calibration example; *CalibrationTutorial* by L. Ferrer.